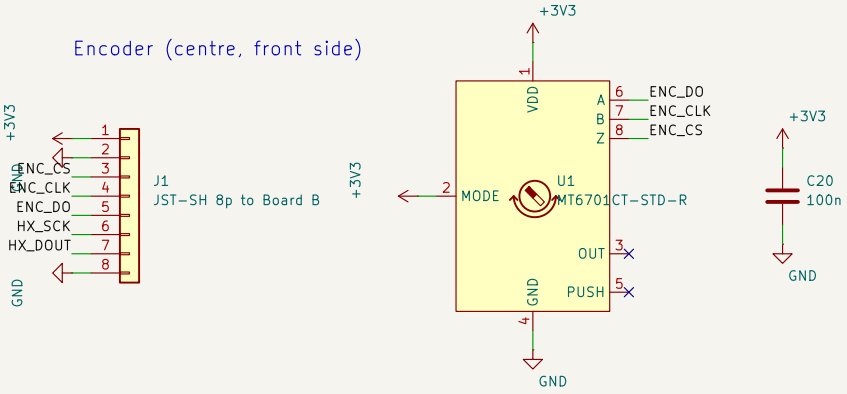
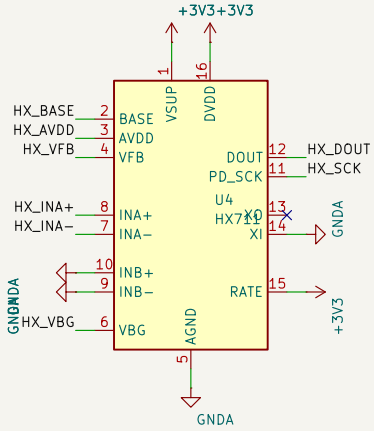
Board A "knob-base" (GaggiMate knob rev 1). Circuit from SmartKnob View view_base (Scott Bezek, Apache-2.0):
kept MT6701 encoder, HX711 + PNP AVDD regulator, 4x 2512 strain-gauge bridge, GNDA net tie.
Removed: ESP32, USB/CH340, regulators, TMC6300, LEDs, VEML7700, LCD pads, J4/J5 gauge pads.
Changes: L1 3.3uH -> FB1 basic ferrite bead; S8550-family PNP (C8542, basic) instead of MMBT4403;
power + data from Board B on J1 (JST-SH 8p): 1 3V3, 2 GND, 3 ENC_CS, 4 ENC_CLK, 5 ENC_DO, 6 HX_SCK, 7 HX_DOUT, 8 GND.

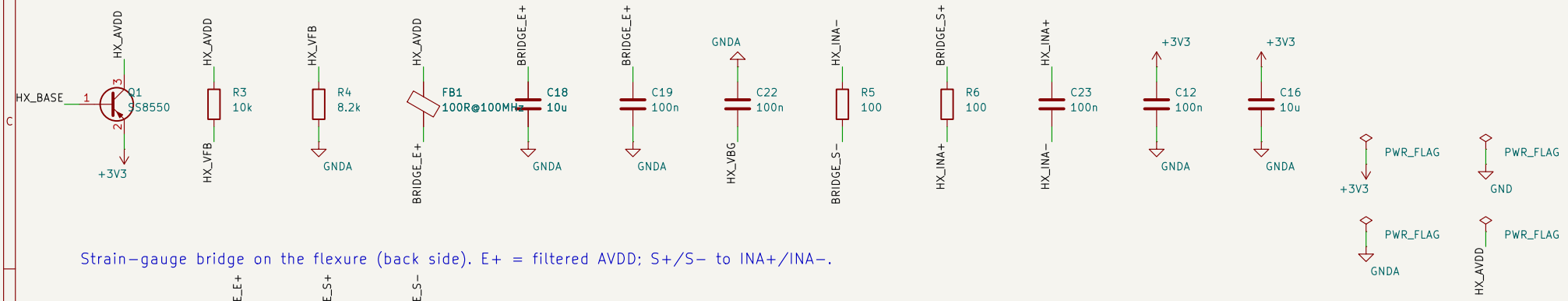
Encoder (centre, front side)



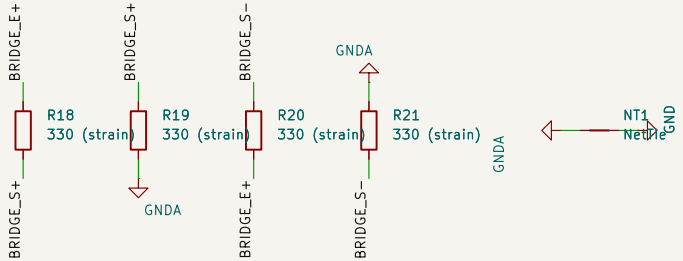
Press sensor ADC



HX711 AVDD regulator, excitation filter, input filter, decoupling



Strain-gauge bridge on the flexure (back side). E+ = filtered AVDD; S+/S- to INA+/INA-.



MT6701 encoder + HX711 strain-gauge press sensor; 3V3 from Board B via JST-SH 8p
Derived from SmartKnob View view_base by Scott Bezek, Apache-2.0

GaggiMate knob (personal fork)

Sheet: /
File: knob-base.kicad_sch

Title: GaggiMate knob — Board A knob-base

Size: A4	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1